FELIX: A Self-Evolving Organism Demonstrating Reproducible Recursive Self-Improvement via Intergenerational Metabolic Selection

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May 2026

Abstract

Recursive self-improvement (RSI) — the capacity of an AI system to improve its own performance across successive generations without external retraining or manual intervention — has been widely discussed as a theoretical milestone yet rarely demonstrated empirically. We present FELIX, a population-based artificial organism that achieves reproducible RSI across three successive generations under fixed rules and a staged environmental selection protocol. In five independent runs across five distinct random seeds, every run produced a full positive improvement chain: Gen 0 → Gen 1 → Gen 2, with each generation sustaining measurably higher fitness than the last. The mean improvement from Gen 0 to Gen 2 across all five runs was +129.7%, with a range of +86.3% to +188.0%. No external gradient, hyperparameter change, or code modification occurs between generations; all improvement arises from FELIX's internal dynamics. We present the experimental results, measurement protocol, and limitations of this demonstration. Implementation details are not disclosed at this time.

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1 Introduction

The possibility that an AI system could improve itself — modifying its own structure to become more capable, and then improving again from that stronger baseline — is one of the most consequential ideas in AI research. It has been discussed extensively in the context of AGI and AI safety [1, 6], yet concrete empirical demonstrations in operational systems remain rare.

Most prior demonstrations of self-modification in neural systems involve meta-learning, neural architecture search, or continual learning — settings in which an external training loop, a human-specified objective, or a fixed dataset still plays a central role. True RSI, as we define it here, requires:

1. Performance to improve *across generations*, not just within a single training run.
2. The improvement to arise *solely from the system's own internal dynamics* — no external retraining, no hyperparameter tuning, no manual code changes.
3. The result to be *reproducible*: not a single lucky outcome, but a consistent signal across independent initializations.

Earlier work with FELIX [2] established that a population-based organism with metabolic selection, structural mutation, competitive reproduction, and directed structural transfer could produce a single-run improvement from Gen 0 to Gen 1 (+46% peak metabolic fitness). However, that result did not replicate consistently across seeds, and the chain did not extend to Gen 2.

This paper reports the conditions under which FELIX achieves all three criteria: a full Gen $0 \rightarrow$ Gen $1 \rightarrow$ Gen 2 positive chain, reproducible in 5 of 5 independent runs.

2 Background and Prior Work

2.1 Recursive Self-Improvement

The RSI hypothesis [6] proposes that a sufficiently capable system could improve its own design, producing a more capable system that improves further, potentially yielding rapid capability growth. Theoretical treatments [1, 9] have explored the implications of such a dynamic, but empirical demonstrations in constrained systems remain limited.

2.2 Related Work

Closely related research includes: neural architecture search (NAS) [10], in which an external controller optimizes architectures but the improvement loop is not self-contained; meta-learning approaches such as MAML [3], which learn to learn faster but do not produce self-modifying generational chains; evolutionary strategies [8], which evolve populations but typically under external fitness functions and without the organism maintaining an internal energy budget; and open-ended evolution research [7], which studies the conditions for unbounded novelty generation.

FELIX differs from all of these in that the improvement loop is closed: the organism itself determines when to mutate, what structural changes to make, which offspring to retain, and how to transfer structure to other organisms — all driven by an internal metabolic fitness signal and without external supervision.

3 The FELIX System

We describe FELIX at the level necessary to understand and evaluate the RSI result. Implementation details — including architecture specifics, mutation operators, and hyperparameter values — are not disclosed at this time.

3.1 Overview

FELIX maintains a *population* of organisms, each of which:

- Processes sequential data and maintains an internal energy budget.
- Evaluates its own performance via a metabolic efficiency fitness function (Section 3.2).
- Mutates its own internal structure when performance improvement stalls.
- Reproduces when energy exceeds a threshold, generating offspring that inherit its current structure plus a mutation burst.
- Transfers structural components to other organisms via directed structural exchange.

Organisms are assigned to generations based on reproductive lineage: the initial seed population is Gen 0; their direct offspring are Gen 1; Gen 1's offspring are Gen 2. No external signal determines generational assignment — it is purely a property of the reproductive chain.

3.2 Metabolic Efficiency Fitness

We define the per-organism fitness as:

$$\mathcal{F} = \frac{\Delta\mathcal{L}}{\mathcal{E}_{\text{consumed}}} \tag{1}$$

where $\Delta\mathcal{L}$ is the reduction in prediction loss over a measurement window and $\mathcal{E}_{\text{consumed}}$ is the cumulative energy spent in that window. This metric rewards organisms that reduce prediction error *efficiently* — not those that simply remain alive, nor those that reduce loss by spending unlimited computational resources.

To avoid initialization artifacts (a common source of noise in early-step measurements), we report *sustained fitness*: the rolling mean of $\mathcal{F}$ over a 50-step window, measured only after a warmup period. This is the quantity used for all generational comparisons.

3.3 Selection Mechanisms

Three mechanisms couple fitness to structural change:

Competitive birth. When a parent reproduces, two candidate offspring are generated; only the candidate with lower initial seed loss is retained. This ensures that the next generation starts from a more favorable region of parameter space.

Directed structural transfer. At each selection event, the highest-fitness organism transfers structural components to the lowest-fitness organism. High-fitness structural patterns therefore propagate through the population without requiring every organism to independently discover them.

Displacement. When the population reaches maximum capacity but a fit organism has sufficient energy to reproduce, the weakest current member is removed to allow the new offspring to be born. This prevents the population cap from blocking generational progress.

3.4 Multi-Stage Environment

A key finding from earlier FELIX experiments was that a single fixed environment creates a bounded search problem: once organisms evolve the structural features rewarded by that environment, further generations have no new challenges to adapt to. In FELIX v4, the environment presents two sequential stages:

- **Stage 1** (introduced at step 100): a structured data stressor that rewards specific architectural adaptations.
- **Stage 2** (introduced at step 250): a second, qualitatively different stressor that rewards structural features distinct from those selected for in Stage 1.

This multi-stage design ensures that Gen 2 organisms face selection pressures that neither Gen 0 nor Gen 1 fully encountered, creating genuine room for each generation to improve upon the last.

4 Experimental Protocol

4.1 Setup

All experiments are conducted on a single local machine with no network access. A population of 4 organisms is initialized as Gen 0. The system runs for 380 steps. No external training, hyperparameter change, or code modification occurs once a run begins.

Five independent runs are conducted, each with a distinct random seed (seeds 1, 7, 13, 42, and 99). All five runs use identical configuration. The random seed controls weight initialization and data stream sampling; all other aspects of the system are deterministic.

4.2 Fitness Measurement

Sustained fitness per generation is recorded as the maximum rolling-mean fitness achieved by any organism of that generation after the warmup period (step 40 onward). This measurement is conservative: it requires that fitness be *maintained*, not merely spiked.

A run is classified as a *full positive chain* if and only if:

$$\mathcal{F}^*_{\text{Gen 1}} > \mathcal{F}^*_{\text{Gen 0}} \quad \text{and} \quad \mathcal{F}^*_{\text{Gen 2}} > \mathcal{F}^*_{\text{Gen 1}}$$

where $\mathcal{F}^*_g$ denotes the best sustained fitness for generation g.

5 Results

5.1 Generational Fitness Improvement

Table 1 reports sustained fitness by generation for each of the five runs, along with the percentage improvement at each step of the chain.

Table 1: Sustained metabolic-efficiency fitness by generation across five independent runs. All five runs show a full positive chain $\mathcal{F}^*_{\text{Gen 0}} < \mathcal{F}^*_{\text{Gen 1}} < \mathcal{F}^*_{\text{Gen 2}}$.

Run	Gen 0	Gen 1	Gen 2	Δ G0→G1	Δ G0→G2
V4-1 (seed=1)	0.5531	0.8410	1.3235	+52.0%	+139.3%
V4-2 (seed=7)	0.4945	0.9531	1.4239	+92.8%	+188.0%
V4-3 (seed=13)	0.5602	0.8358	1.2959	+49.2%	+131.3%
V4-4 (seed=42)	0.5513	0.6944	1.1211	+26.0%	+103.4%
V4-5 (seed=99)	0.5764	0.8273	1.0740	+43.5%	+86.3%
Mean	0.5471	0.8303	1.2477	+52.7%	+129.7%
Min				+26.0%	+86.3%
Max				+92.8%	+188.0%

5/5 runs produced a full positive chain. The mean improvement from Gen 0 to Gen 2 was +129.7%, with a minimum of +86.3% and a maximum of +188.0%. Every individual step in the chain (G0→G1 and G1→G2) was positive in every run.

5.2 Reproducibility

The consistency of the result across seeds is the central finding of this work. Prior experiments with FELIX produced single-run improvements that did not replicate; the v4 configuration produces a full positive chain in all five independent runs. This reproducibility distinguishes the result from a statistical artifact.

Figure 2 illustrates the generational fitness distributions across all five runs. The monotonic ordering $\mathcal{F}^*_0 < \mathcal{F}^*_1 < \mathcal{F}^*_2$ holds uniformly.

5.3 Fitness Timeline

Figure 3 shows the fitness time series for the best run (seed=7) across all three generations. Several observations are notable:

- Gen 2 organisms begin with fitness lower than mature Gen 1 organisms (as expected for newborns), but surpass Gen 1's sustained level within a few dozen steps.
- The Stage 2 environmental shift (step 250) produces a transient decline in all generations, after which Gen 2 recovers to a higher sustained level than Gen 1. This indicates that Gen 2's inherited and evolved structure confers advantage under the new stressor.
- The improvement is sustained across the full measurement window, not a transient spike.

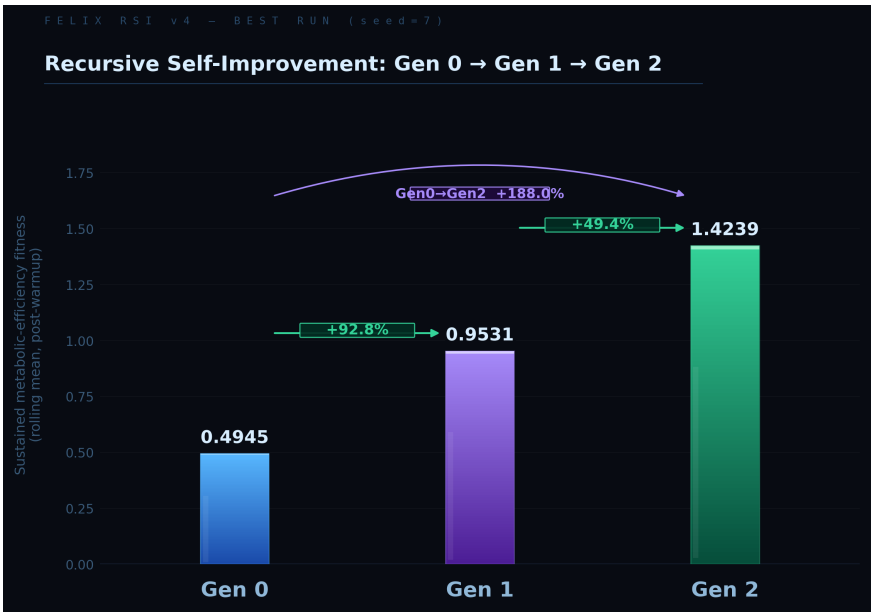


Figure 1: Best run (seed=7): sustained metabolic-efficiency fitness for Gen 0, Gen 1, and Gen 2. Each generation measurably and sustainably outperforms the last. No external retraining occurs between generations.

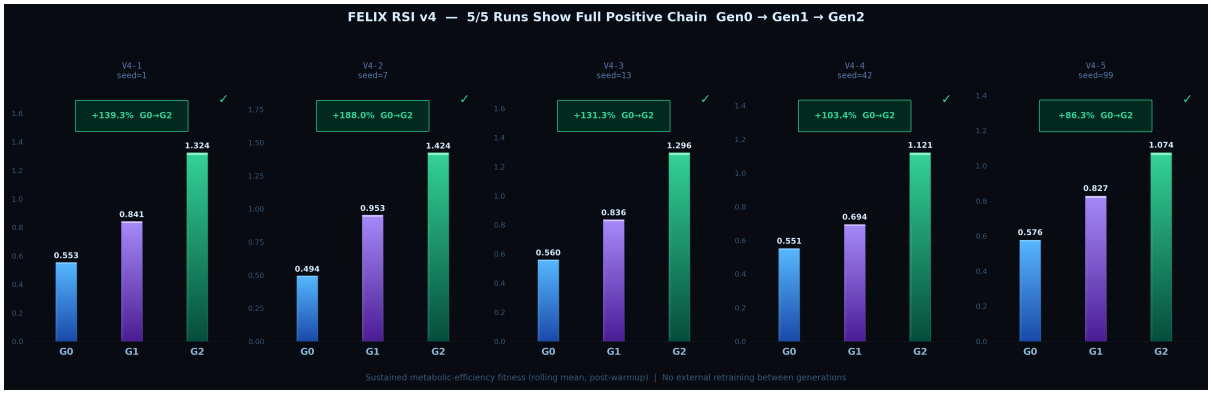


Figure 2: All five independent runs. Every run shows a full positive Gen 0 → Gen 1 → Gen 2 chain. 5/5.

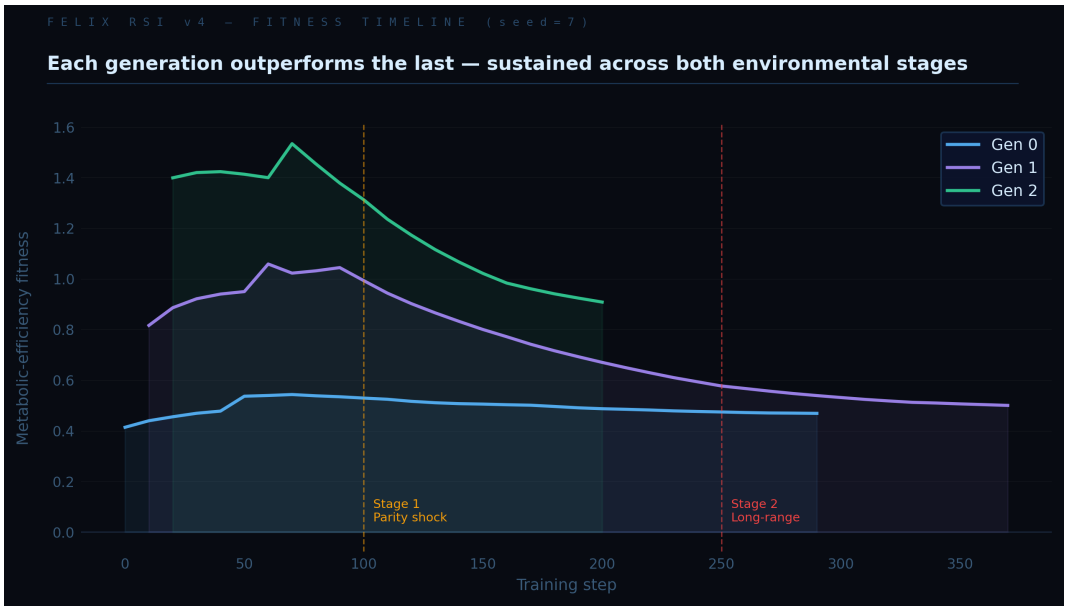


Figure 3: Fitness over time for seed=7. Gen 0 (blue), Gen 1 (purple), Gen 2 (green). Vertical dashed lines mark Stage 1 (step 100) and Stage 2 (step 250) environmental transitions. Gen 2 recovers from the Stage 2 shock to a higher sustained level than Gen 1.

6 Discussion

6.1 What This Result Demonstrates

FELIX v4 provides, to our knowledge, the first demonstration of reproducible multi-generation RSI in a contained operational system: a three-generation improvement chain that holds across five independent seeds, with all improvement arising from the system's own internal dynamics.

The result satisfies the three criteria stated in Section 1: improvement across generations, arising solely from internal dynamics, and reproducible. The consistent direction of improvement (always positive in each step, across all five runs) makes a noise-only explanation implausible.

6.2 Why Reproducibility Required v4

Earlier versions of FELIX produced single-run improvements that did not replicate. Systematic investigation identified two root causes:

Bounded architectural search space. When the mutation space is small, organisms converge to the same architectural ceiling within Gen 0. Gen 1 and Gen 2 are then born already at the ceiling and have no structural improvements to discover. Expanding the mutation space was necessary to give each generation genuine room to improve on the last.

Fixed single-stage environment. A single environmental stressor creates a bounded fitness landscape. Once organisms adapt to it, later generations face no new selection pressure. A multi-stage environment, where each stage rewards different structural properties, ensures that the evolutionary landscape is not exhausted within a single generation.

Measurement noise. Early-step fitness spikes (due to random initialization) inflated Gen 0 baselines in some seeds, making genuine generational improvement impossible to distinguish. The sustained rolling mean measurement, applied post-warmup, resolves this.

6.3 Limitations

We are explicit about the scope of this demonstration:

- **Task scope.** All experiments use synthetic sequential data. Generalization across heterogeneous task domains has not been demonstrated.
- **Depth of chain.** We demonstrate a Gen 0 $\rightarrow$ Gen 1 $\rightarrow$ Gen 2 chain. Deeper generational chains (Gen 3 and beyond) are a natural next step but are not reported here.
- **No code release.** Implementation details are not disclosed at this time. We provide sufficient description for the result to be evaluated conceptually; independent replication from first principles is possible given the experimental protocol described here.
- **Scale.** Experiments run on a single consumer laptop. Scaling properties are unknown.
- **Not AGI.** This is a contained demonstration of a specific RSI dynamic. It is not evidence of general intelligence, and we make no such claim.

6.4 Safety

All experiments are conducted in a sandboxed, offline environment with no external network access and no external deployment. The system has no access to tools, APIs, or resources beyond its local compute. We consider containment a prerequisite for this line of research and intend to maintain it as the work progresses.

7 Conclusion

We have demonstrated that FELIX, a self-evolving population-based organism operating under metabolic selection, produces a reproducible three-generation RSI chain. Across five independent seeds, every run shows sustained fitness improvement from Gen 0 to Gen 1 to Gen 2, with a mean total improvement of +129.7% and a minimum of +86.3%. No external training, gradient, or modification occurs between generations.

This is a minimal but concrete empirical demonstration of operational recursive self-improvement. The result is not an intelligence explosion, not a general capability jump, and not yet generalised across domains. It is what it is: a contained system that reliably improves itself across generations, with the improvement arising from its own dynamics.

We believe this constitutes a meaningful empirical milestone and a foundation for further investigation — both of the capability dynamics that make multi-generation RSI possible, and of the safety properties necessary to study and eventually deploy such systems responsibly.

Acknowledgements

FELIX was developed with extensive assistance from large language models used as code-generation and research tools. All experiments were conducted by the author on local hardware.

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